

overall system performance, instead of the traditional single-core processor designs. With multiple cores on a single chip executing simultaneously, it is possible to meet the throughput demands of an application while keeping the processor clock frequencies moderate, to contain power consumption and the heat generated. So, what do multi-core processors mean to future database systems?

Multi-core systems require database systems that can extract greater parallelism, in both intra- and inter-query processing. This requires Massively Parallel Databases (MPP Databases), which support 'Shared Nothing' architectures. While parallel databases are known to scale to hundreds of nodes, they have not been shown to scale to thousands of nodes and beyond. On the other hand, non-traditional database architectures, like data stores supporting Map-Reduce, have been scaled to thousands of nodes with extremely high availability and fault resilience. Hence, the question that needs to be answered is: Are massively parallel databases the right solution for tomorrow's multi-core era or is it the Map-Reduce data stores? A detailed discussion that compares the relative merits of these two systems was described in the paper, 'A Comparison of Approaches to Large-Scale Data Analysis', available at <http://www.cse.nd.edu/~dthain/courses/cse40771/spring2010/benchmarks-sigmod09.pdf>.

Current state-of-the-art approaches have also proposed combining massively parallel databases with the Hadoop architecture, as proposed in HadoopDB, which is described in the paper, 'HadoopDB: An Architectural

Hybrid of MapReduce and DBMS Technologies for Analytical Workloads' available at <http://db.cs.yale.edu/hadoopdb/hadoopdb.pdf>.

Another interesting direction for the future of database technology is to take advantage of the emerging GPU Many-Core architectures. There has been work on off-loading many of the data query operators to GPU architectures. Techniques for mapping many database operations to GPUs are discussed in the paper, 'Fast Computation of Database Operations using Graphics Processors', available at <http://www.cs.unc.edu/~lin/db.pdf>.

The most important hardware innovation for future databases is, of course, the introduction of Non-Volatile RAM. Databases have traditionally been using the buffer pool in memory to speed up database operations, while also maintaining the ACID properties. The introduction of NVRAM takes away the need for the buffer pool, and requires a database re-design. While current databases have been designed for optimising disk reads, which are much slower than main-memory accesses, this will change drastically with NVRAM, since RAM is going to be persistent. Therefore, newer database systems explicitly designed for NVRAM are the need of the hour.

While we have been discussing the challenges for future databases, I would like to bring to the reader's attention an informative website on state-of-the-art database design: dbmsmusings.blogspot.com/, which is maintained by Prof Daniel Abadi, who is a pioneer in modern database design. His insightful articles on database research are a must-read for database programmers. 

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